

ISELED/ILaS Example using E²B

User Guide
Version 2.0

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1. Introduction

This document provides instructions to run the ISELED (Intelligent Smart Embedded LED) and ILaS (ISELED Light and Sensor) using E²B (Ethernet to the Edge Bus) devices, hardware setup details and software setup details are discussed.

- **AD3304:** Supports up to 2 ISELED interfaces.
- **AD3305:** Supports up to 4 ISELED interfaces.

Note: To use the ISELED or ILaS interface on E²B devices (AD3304/05), NDA from Inova Semiconductors is mandatory.

1.1 Terminology

To ensure a clear understanding of the content in this document, it is helpful to first become familiar with the following terms.

Table 1: Abbreviations, Terms and Symbols

Acronym	Description
ADK*	Application Development Kit
E ² B	Ethernet to the Edge Bus
ESP*	Extender Side Port
GND	Ground
ILaS	ISELED Light and Sensor
ISELED	Intelligent Smart Embedded LED
ISP*	Initiator Side Port
LED	Light Emitting Diode
RGB	Red, Green and Blue

*The terminology is from Inova INLT220Q user manual.

Difference between ISELED and ILaS can be found here: <https://inova-semiconductors.de/iseled.html>

2. Revision History

Table 2: Revision History

Version	Description
1.0	Initial Release
2.0	Updated for EdgeStudio 3.5.1 release • ILaS connection diagram • Bench setup picture • EdgeStudio quick start guide section names.

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3. Hardware

The section provides information about the hardware components required to run ISELED and ILaS with E²B devices.

3.1 E²B Hardware

The following hardware components are required to set up and run the ISELED example with E²B devices.

- 1x E²B AD3300/6 Controller Node Board
- 1x E²B AD3304/5 Remote Node Board
- 1x E²B AD330x Remote Board Breakout Board

Please refer to the documents below to learn more about the above E²B Devices

- E2B_Customer_Demo_User_Guide_Rev1p9.pdf
- AD330x_Customer_Demo_Adaptor_Board_User_Guide_v1_0.pdf

3.2 ISELED Hardware

The ISELED hardware below is required to run the `ex_iseled` example.

3.2.1 ISELED Board^[1]

The ISELED board contains 16 calibrated SmartRGB LEDs from Dominant Opto Technologies, two blocking capacitors (100nF, 1μF) at each device, a 7V to 5V low dropout regulator and two 8 position nanoMQS headers from Tyco Electronics. Each RGB package includes an INLC100D controller chip from Inova Semiconductors with which the devices are calibrated to the same color gamut and brightness.

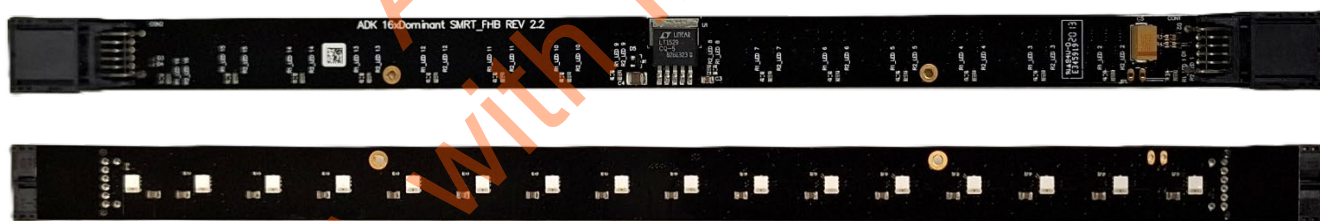


Figure 1 ADK 16xDominant SMRT_FHB Rev2.2 (Primary and Secondary side)

3.2.2 ISELED Power Adapter Board^[2]

The power adapter board contains a DC power jack to supply the LEDs. Besides, there are two DC/DC converters (12V to 7V and 7V to 5V) and an INLC100E device. This power adapter board provides the INLC100E LED driver chip which converts the single ended communication signal between E²B and the device into a differential output signal.

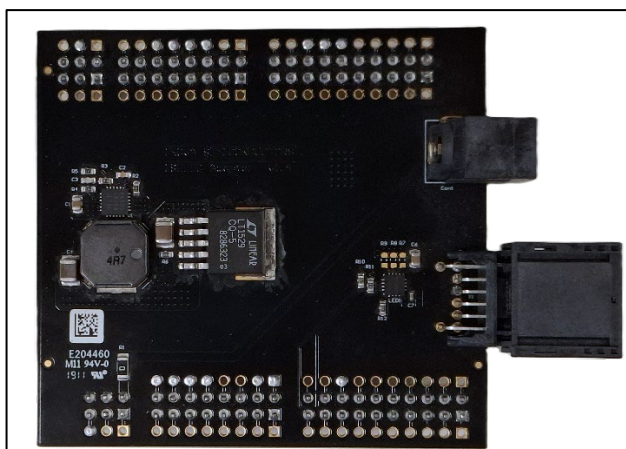


Figure 2 ISELED Adapter Version 1.4

3.2.3 Board Connection Cable

The 20 cm custom eight-wire cable includes two twisted wires dedicated to serial differential communication, along with three wires each for 7V power and ground. The wire ends are crimped and connected to NanoMQS connectors from TE Connectivity (Tyco Electronics).



Figure 3 Board Connection Cable

3.3 ILaS Hardware

The ILaS hardware below is required to set up the ILaS with E²B.

3.3.1 ILaS Board^[3]

The ILaS Board is an evaluation and development platform designed to demonstrate and test the capabilities of the INLT220Q transceiver, which is part of Inova's ILaS (ISELED Light and Sensor) network technology.

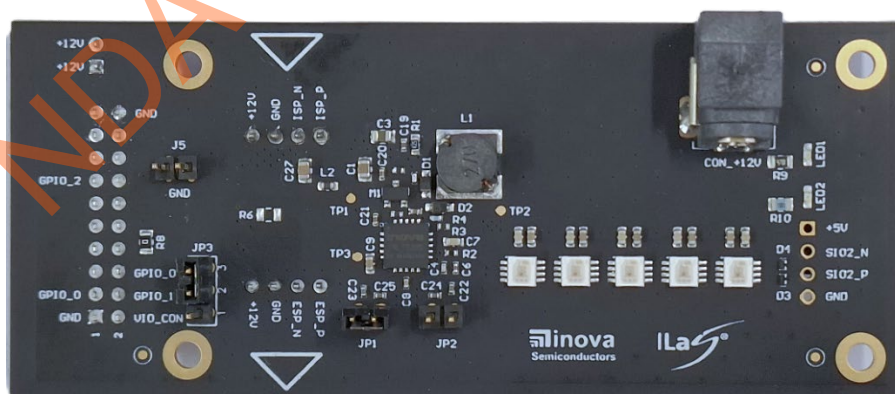


Figure 4 INLT220Q-ADK

4. E²B Hardware Setup (Common for ISELED & ILaS)

The typical hardware setup which is common for ISELED and ILaS using SC594 SOM as Host:

1. Set up the E²B Hardware by following the instructions mentioned in the E²B Customer Demo User guide^[4]. Contact ADI team to get this document.
2. Hardware connections on E²B AD3304/5 Remote Node Board
 - Connect the following jumpers on E²B AD3304/5 Remote Node Board.
 - JP1: 2&4
 - P5: 1&2 and 3&4 (Termination)
 - P1: Can be ignored for this example.
 - Connect one end of 10BASE-T1S cable harness to 'P3' of E²B AD3304/5 Remote Node Board.
 - Connect the power cable to 'P8' of the E²B AD3304/5 Remote Node Board.
3. Hardware Connections on AD3300/6 E²B AD3300/6 Controller Node Board
 - Switch the S_MUX switch to position '5' (SOM) to make SC594 SOM as Host.
 - Connect the other end of 10BASE-T1S cable harness to 'ECONN1' or 'ECONN2' port of the Controller board.
 - Connect the USB type A to C cable from PC to 'P3' of SC594 SOM. This is to see UART logs in the console and for flashing loader file.
 - Observe the setup as shown (without ISELED or ILaS Hardware).

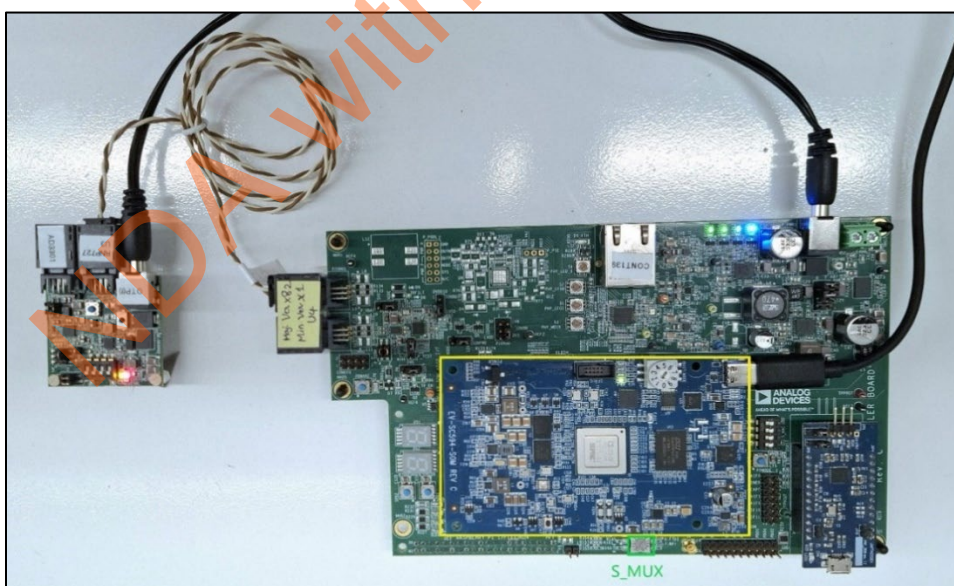
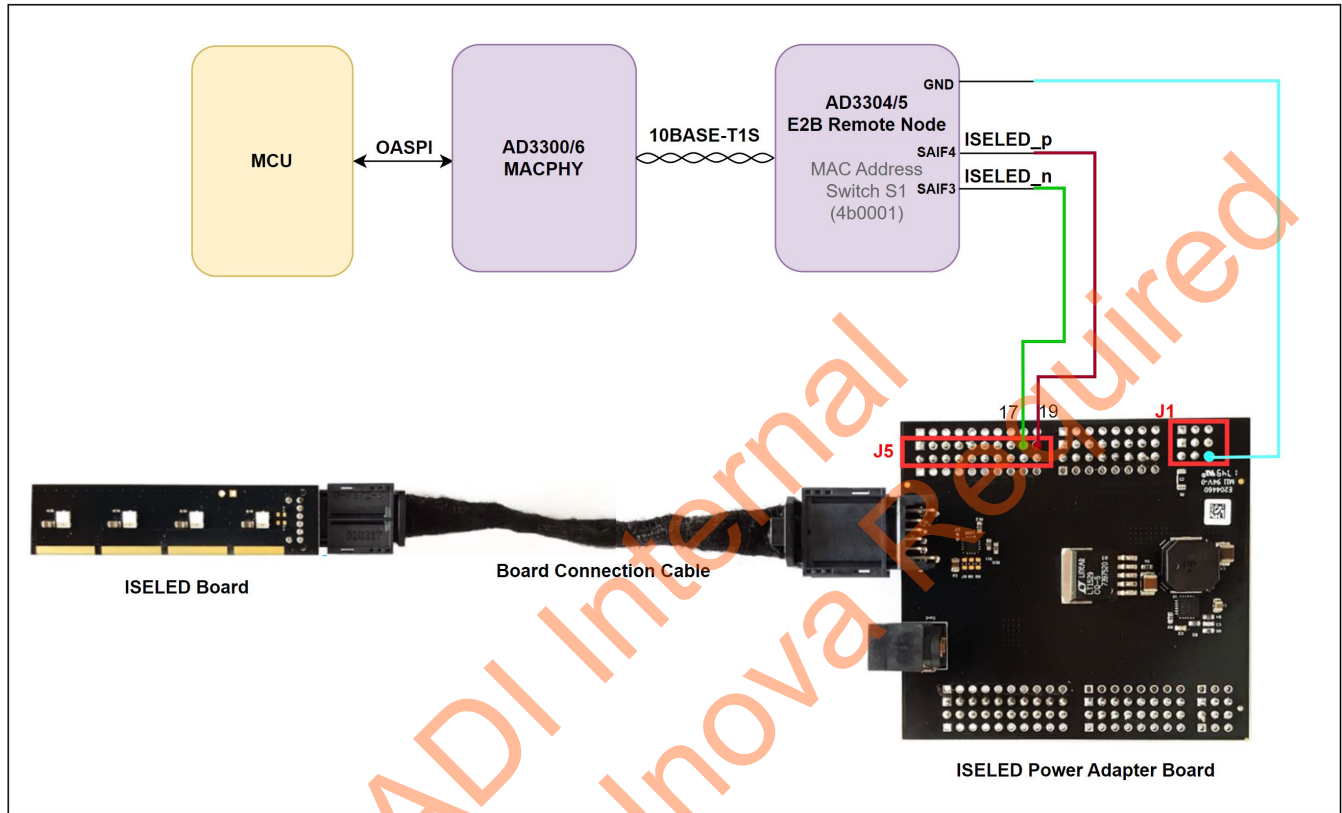


Figure 5 E²B P2P Setup (Common setup for ILaS and ISELED)

5. ISELED with E²B

5.1 E²B - ISELED Hardware Setup Block Diagram



Note: ISELED Board image has been cropped to fit in the block diagram.

Figure 6 E²B - ISELED Hardware setup

5.2 Hardware Setup

To make setup, connect all components in the following order:

- Connect the E²B hardware setup as mentioned in section “4. E2B Hardware Setup (Common for ISELED & ILaS)”.
- Connect one end of the board connection cable to X1(nanoMQS port) of ISELED power adapter board.

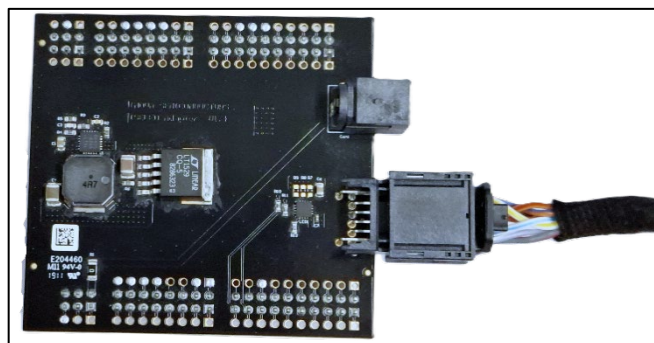


Figure 7 ISELED Power Adapter to Board Connection Cable

- Connect the other end of the board connection cable to CON1(Nano MQs port) of ISELED board as shown.

Note: Do not connect to CON2, it can be used to connect another branch of ISELED LED strip.

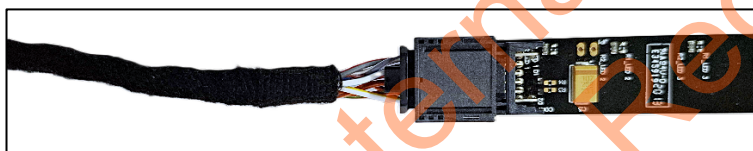


Figure 8 ISELED Board to Board Connection Cable

- Connect the SAIF3 (ISELED_n) to pin-17 of J5 and connect the SAIF4 (ISELED_p) to pin-19 of J5 on ISELED power adapter board as shown in the “E²B-ISELED Hardware Setup Block Diagram” and Connect Ground (GND) between AD330x Remote Board Breakout Board^[6] and ISELED Power adapter board.
- Connect the power supply to Con1 of ISELED Power Adapter Board.
- Connect the power supply to all E²B hardware and power on setup.

5.3 Software

To run the example, follow the steps below.

5.3.1 Run the Example

1. Copy the *ex_iseled* example application to *construction/app* folder

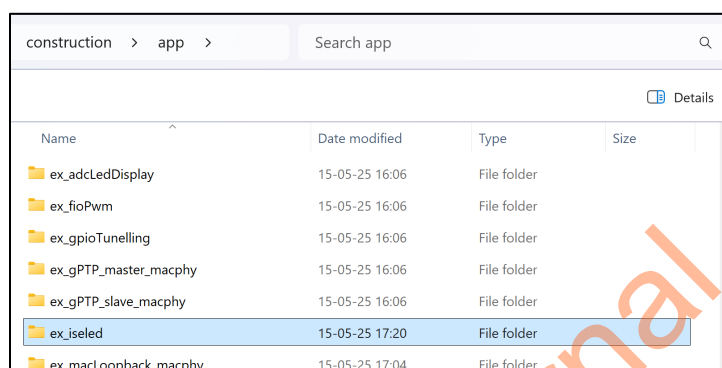


Figure 9 *ex_iseled* copying to *sdk* folder

2. Go to the file *construction/app/CMakeLists.txt* and add the line below.
"add_subdirectory(ex_iseled)"

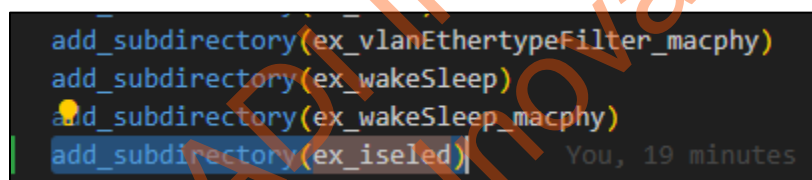


Figure 10 Modifying CMake file for building executable

3. Build the application by following the steps provided in chapter 6. *Building Example applications* in EdgeStudio Quick start guide^[5]. (Path: C:\Analog Devices\EdgeStudio x.y.z\docs\QuickStartGuide).
4. Run the example by following the steps provided in chapter 7. *Running Example Applications* in EdgeStudio Quick start guide^[5]. (Path: C:\Analog Devices\EdgeStudio x.y.z\docs\QuickStartGuide).
5. To see the terminal logs, setup UART Terminal by following the instructions mentioned in the section "4.2 UART Terminal" of EdgeStudio Quick start guide^[5]. The output console can be seen in Figure 11.

```

=====
Example - ISELED
=====
- Periodically updates the ISELED LEDs.
- Interfaces used: ISELED.
=====
>>>> App build time : May 15 2025 18:00:15
>>>> E2bilib build time: May 15 2025 18:00:04
>>>> E2B Stack version: 3.5.0

>>>> Host name: HOST_SOM_SC594
>>>> Os type: FreeRTOS
>>>> Initialization sequence started
>>>> Platform Abstraction Layer initialized
>>>> Network layer initialized
>>>> Got the list of network devices
      Id: 0, Desc: SOM SPI0
      Id: 1, Desc: SOM SPI1
      Id: 0, Desc: Ezkit Eth0
>>>> Opened the network device: Id: 1, Desc: SOM SPI1
      MAC address discovery for instance 0: 00:e0:22:fe:70:01
>>>> Remote1 Major Revision Number: 0x00000083, Minor Revision Number: 0x00000000
ISELED write Response (ID = 0x01)
ISELED write (ID = 0x02)
>>>> Address      Instruction      Data
      0x11      0x00      0x00 0x51 0x07
      0x10      0x00      0x00 0x51 0x03
      0x0f      0x00      0x00 0x51 0x05
      0x0e      0x00      0x00 0x51 0x06
      0x0d      0x00      0x00 0x51 0x0f
      0x0c      0x00      0x00 0x51 0x0c
      0x0b      0x00      0x00 0x51 0x00
      0x0a      0x00      0x00 0x51 0x0b
      0x09      0x00      0x00 0x51 0x04
      0x08      0x00      0x00 0x51 0x07
      0x07      0x00      0x00 0x51 0x09
      0x06      0x00      0x00 0x51 0x05
      0x05      0x00      0x00 0x51 0x0a
      0x04      0x00      0x00 0x51 0x02
      0x03      0x00      0x00 0x51 0x09
      0x02      0x00      0x00 0x51 0x0d
      0x01      0x00      0x00 0x77 0x03

>>>>
>>>> List of Supported Commands:
>>>> 'Ctrl + R' to toggle between command mode and run mode
>>>> Type any of the following when in command mode:
>>>> help          Prints list of supported application commands
>>>> controller_stat Reads controller node MAC and PHY status
>>>> ping          Ping Remote Nodes
>>>> remote_stat   Reads remote nodes' MAC and PHY status
>>>> intf_stat     Reads remote nodes' Interface status
>>>>
Switching to run mode

```

Figure 11 Output Console for ISELED Example

5.3.2 Software Flow for ISELED Driver

1. Reset the ISELED strip
2. Initialize the ISELED, will get response from all the LEDs (INLC100D) present on the ISELED strip. In this case 17 responses (16 LEDs + INLC100E).
3. Send ISELED commands to set RGB colors on the ISELED strip.

5.4 Bench Setup Picture for ISELED



Figure 12 Bench setup - Driving ISELED with E²B, SC594 SOM as host

6. ILaS with E²B

6.1 E²B – ILaS Hardware Setup Block Diagram

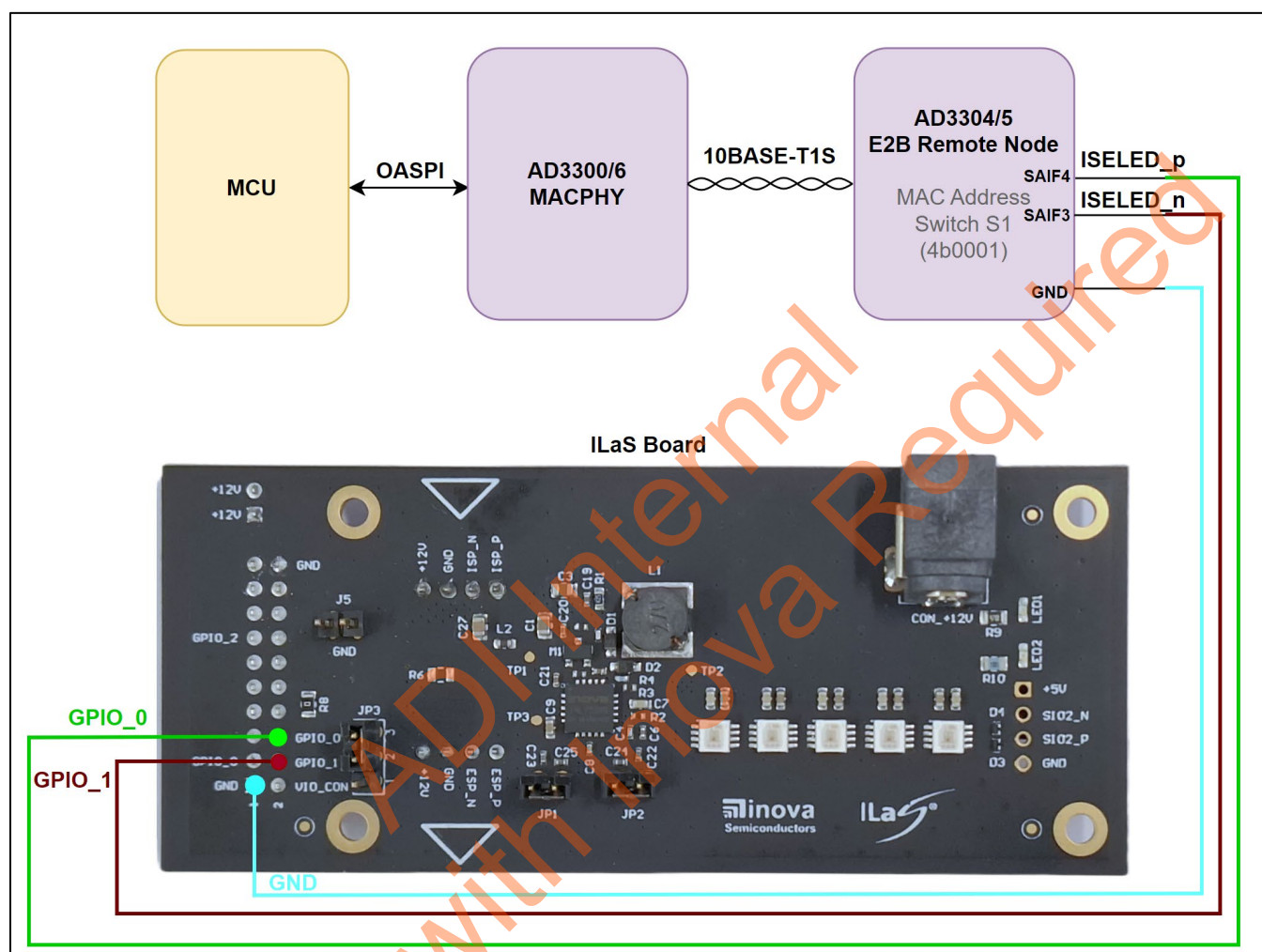


Figure 13 E²B ILaS Block Diagram

6.2 Hardware setup

1. Setup the E²B Hardware by following the instructions “4. E²B Hardware Setup (Common for ISELED & ILaS)”.
2. Connect SAIF3 from remote breakout board to GPIO_1 of ILaS Board.
3. Connect SAIF4 from remote breakout board to GPIO_0 of ILaS Board and connect common for ground (GND) for both boards.
4. Short the ISP_N & ISP_P by placing a jumper to JP1 at position 1-2 and short ESP_P & ESP_N by placing a jumper to JP2 at position at 1-2 on ILaS Board as shown in the block diagram above. These settings are as per the INLT220Q device manual^[8] from Inova semiconductors.

5. Connect power supply to the ILaS board and E²B Hardware.

6.3 Software

To run the example, follow the steps below.

6.3.1 Run the Example

1. Copy the *ex_iseled* example application to *construction/app* folder.

Note: The *ex_iseled* example application can be used to run control ILaS devices with modifications mentioned in the following sections.

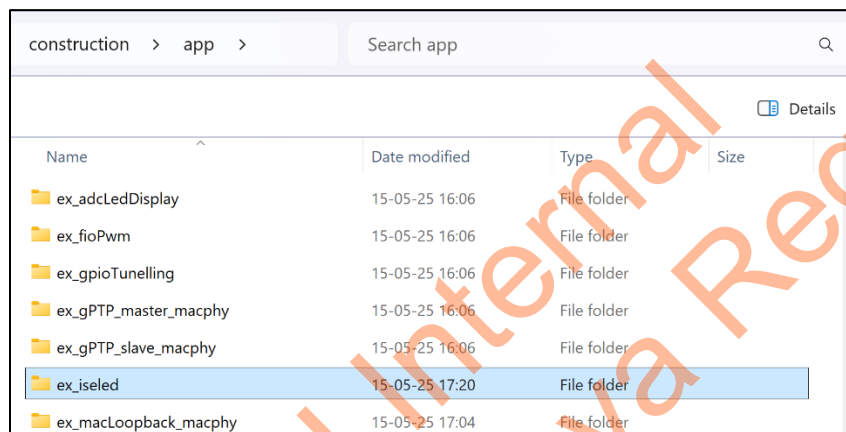
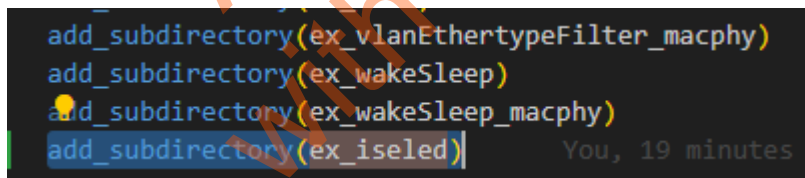


Figure 14 Copying *ex_iseled* to *sdk* folder

2. Go to the file *construction/app/CMakeLists.txt* and add the line *add_subdirectory(ex_iseled)*.
" *add_subdirectory(ex_iseled)* "



```
add_subdirectory(ex_vlanEtherTypeFilter_macphy)
add_subdirectory(ex_wakeSleep)
add_subdirectory(ex_wakeSleep_macphy)
add_subdirectory(ex_iseled)
```

Figure 15 Modifying CMake file for building executable

3. Open the *ex_iseled.e2bprj*, go to settings and select *Iseled1* to change the interface mode from ISELED to ILaS.

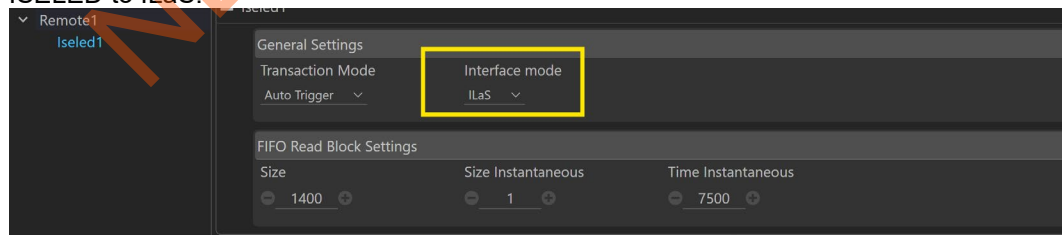


Figure 16 Changing interface mode to ILaS

4. Export the configuration to *ex_iseled* folder.

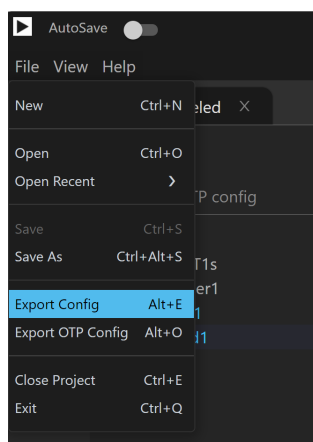


Figure 17 Exporting the configuration for ILaS

5. In `IseledInitStrip` function of file `iseled.c` change the data in `anData` array (`anData[2] = 0x15`) as shown. This is the address of ILaS device.

```
ADI_MEM_CODE_CRIT
ADI_EAL_STATUS IseledInitStrip(uint8_t nRemoteNum)
{
    ADI_EAL_STATUS eEalStatus;
    uint8_t anData[3U] = {0x00u, 0x00u, 0x15u};
    memcpy(ganData[0], anData, 3u);
}
```

Figure 18 Changing the ILaS instruction

6. Change the `nExpectedNumResp` to 6, since our ILaS device has five LEDs and ILaS device.

```
for(uint8_t i = 0u; i < 6u; i++)
{
    aoFastModeData[i].pData = &ganData[i][0u];
}

ADI_E2BCORE_ISELED_FAST_TXN_PAYLOAD *apIseledResp[] = {&aoFastModeData[0u]};

ADI_EAL_ISELED_IO_DATA oIseledIOData;
oIseledIOData.eMode = eMode;
oIseledIOData.nNumIseledCmds = 1U;
oIseledIOData.pIseledCmds = aoFastModeData;
oIseledIOData.nExpectedNumResp = 6U;
oIseledIOData.apIseledResps = apIseledResp;
```

Figure 19 Changing the Expected responses

7. Build the application by following the steps provided in chapter 6. *Building Example applications* in EdgeStudio Quick start guide^[5]. (Path: `C:\Analog Devices\EdgeStudio x.y.z\docs\QuickStartGuide`).
8. Run the example by following the steps provided in chapter 7. *Running Example Applications* in EdgeStudio Quick start guide^[5]. (Path: `C:\Analog Devices\EdgeStudio x.y.z\docs\QuickStartGuide`).
9. To see the terminal logs, setup UART Terminal by following the instructions mentioned in the section 4.2 *UART Terminal* of EdgeStudio Quick start guide^[5]. The output console can be seen in Figure 20.


```
=====
Example - ISELED
- Periodically updates the ISELED LEDs.
- Interfaces used: ISELED.
=====
>>>> App build time : Oct 15 2025 18:19:32
>>>> E2bilib build time: Oct 15 2025 16:30:09
>>>> E2B Stack version: 3.5.1

>>>> Host name: HOST_WIN_X64
>>>> Os type: NoOS
>>>> Initialization sequence started
>>>> Platform Abstraction Layer initialized
>>>> Network layer initialized
>>>> Got the list of network devices
      Id: 0, Desc: FT4222_Loc21131
      Id: 0, Desc: Network adapter 'WAN Miniport (Network Monitor)' on local host
      Id: 1, Desc: Network adapter 'WAN Miniport (IPv6)' on local host
      Id: 2, Desc: Network adapter 'WAN Miniport (IP)' on local host
      Id: 3, Desc: Network adapter 'Intel(R) Wi-Fi 6E AX211 160MHz' on local host
      Id: 4, Desc: Network adapter 'Microsoft Wi-Fi Direct Virtual Adapter #2' on local host
      Id: 5, Desc: Network adapter 'Microsoft Wi-Fi Direct Virtual Adapter' on local host
      Id: 6, Desc: Network adapter 'Adapter for loopback traffic capture' on local host
      Id: 7, Desc: Network adapter 'Realtek USB GbE Family Controller #3' on local host
      Id: 8, Desc: Network adapter 'SonicWall VPN Adapter' on local host
>>>> Opened the network device: Id: 0, Desc: FT4222_Loc21131
      MAC address discovery for instance 0: 00:e0:22:fe:70:01
>>>> Remote1 Major Revision Number: 0x00000083, Minor Revision Number: 0x00000000
ISELED write Response (ID = 0x01)
ISELED write (ID = 0x02)
>>>> Address      Instruction      Data
      0x06      0x00          0x00 0xd1 0x08
      0x05      0x00          0x00 0xd1 0x07
      0x04      0x00          0x00 0xc1 0x05
      0x03      0x00          0x00 0xc1 0x0e
      0x02      0x00          0x00 0xc1 0x0a
      0x01      0x00          0x02 0x07 0x0d

>>>> -----
>>>> List of Supported Commands:
>>>> 'Ctrl + R' to toggle between command mode and run mode
>>>> Type any of the following when in command mode:
>>>> help          Prints list of supported application commands
>>>> controller_stat Reads controller node MAC and PHY status
>>>> ping          Ping Remote Nodes
>>>> remote_stat   Reads remote nodes' MAC and PHY status
>>>> intf_stat     Reads remote nodes' interface status
>>>> -----
>>>>
```

Figure 20 Output Console for ISELED Example

6.3.2 Software flow for ILaS Driver

1. Reset the ILaS board.
2. Initialize the ILaS, will get response from all the LEDs (INLC100D) present on the ISELED strip. In this case 6 responses (5 LEDs (INLC100D) + INLC220Q).
3. Send ISELED commands to set RGB colors on the ILaS strip.

6.4 Bench Setup Picture for ILaS

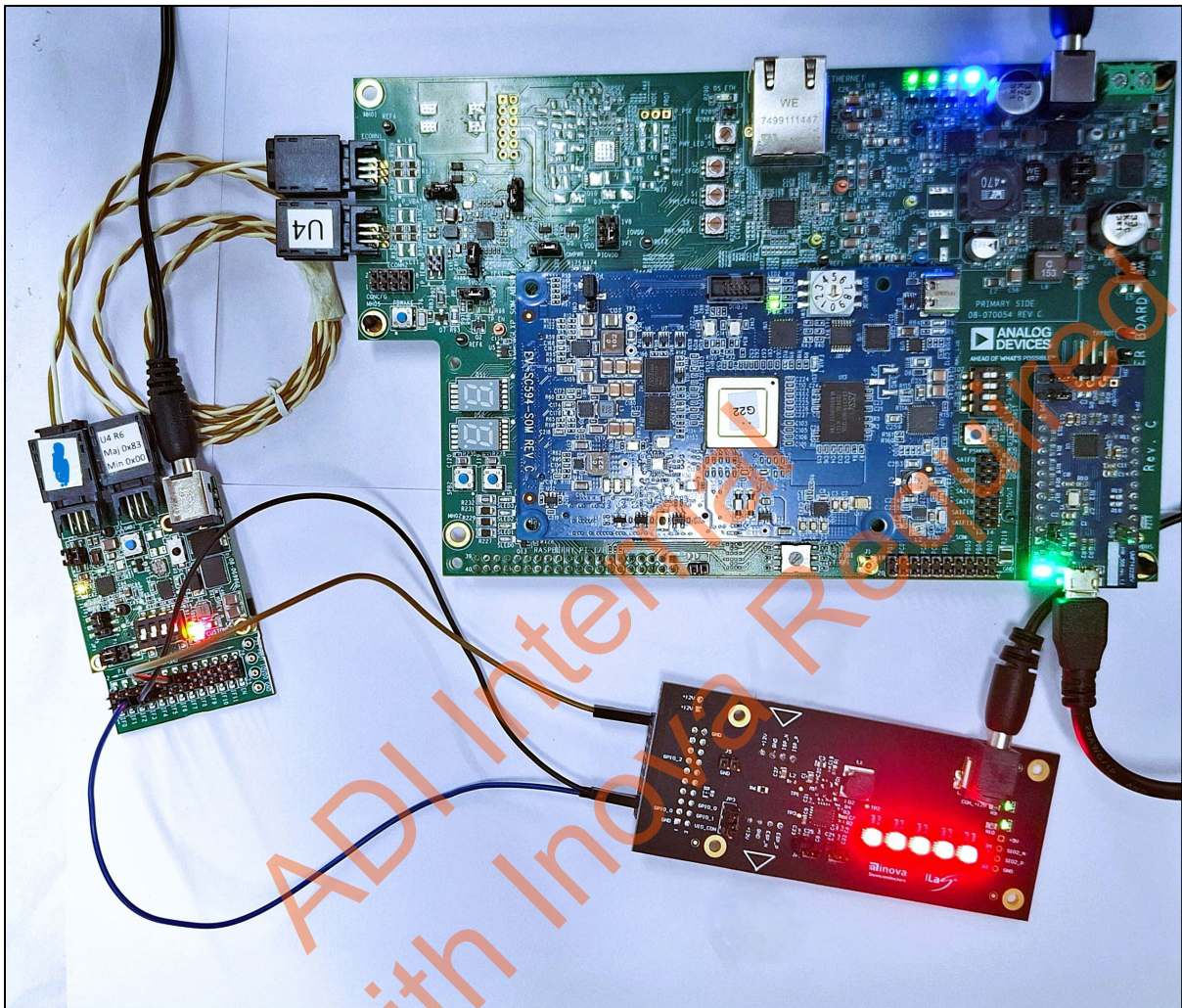


Figure 21 Bench setup - Driving ILaS with E²B

7. References

- [1] ADK 16xDominant SMRT_FHB Rev 2.2
- [2] ISELED Adapter Version 1.4
- [3] INLT220Q_ADK Rev1.3
- [4] E2B Customer Demo User Guide. Rev1.9. Analog Devices, Inc.
- [5] AE_56_EdgeStudio_QuickStartGuide.pdf
- [6] AD330x Customer Demo Adaptor Board User Guide v1.0 Analog Devices, Inc.
- [7] UM_INLC_ADK Revision 0.9_A.pdf
- [8] DS_INLT220Q_Rev 1.0

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